

Virtual Herbal Garden

An Interactive Herbal Encyclopedia

Day WSIE Project Tracker Report

Batch	III AIML_A – 2028
Section	A
Tracking Period	June 01, 2026 – June 13, 2026
Project Duration	June 01, 2026 – June 30, 2026
Subject	Web & Software Integration Engineering (WSIE)
Technology Stack	HTML / CSS / React.js / Node.js / MySQL / Python

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DAY 1 **Activity:** Project Title Finalization | **Deliverable:** Approved Project Title | *Jun-01*

What I Did

On Day 1, I brainstormed and finalised the project title. I explored various ideas related to herbal science, digital education, and web-based applications and selected a topic that combines botanical knowledge with interactive technology to serve students and the general public.

Activities Breakdown

- Brainstormed 10+ project ideas focused on health, education and digital gardening
- Evaluated each idea on feasibility, innovation, and academic value
- Shortlisted 3 topics: Smart Agriculture App, Online Pharmacy, Virtual Herbal Garden
- Finalised title: 'Virtual Herbal Garden – An Interactive Herbal Encyclopedia'
- Obtained faculty approval for the title

Why This Topic?

Reason	Explanation
Real-World Impact	Digitises herbal knowledge for students, healthcare workers, and the general public
Technology Combination	Combines dynamic plant database + interactive UI + search & filtering
Educational Relevance	Supports Ayurveda, ethnobotany, and integrative medicine curricula
Academic Rigor	Covers all required modules: DB, Frontend, Backend, Search, Media, and Security

DAY 2 **Activity:** Requirement Gathering | **Deliverable:** Problem Statement | *Jun-02***Problem Statement**

Traditional herbal knowledge is scattered across books, manuscripts, and oral traditions, making it difficult for students, practitioners, and the general public to access reliable herbal information quickly. There is a need for a centralised, interactive digital platform where users can explore, search, and learn about medicinal herbs, their properties, uses, and cultivation methods in an engaging and structured manner.

Stakeholder Analysis

Stakeholder	Requirements
Students / General Users	Browse and search herbs, view images, medicinal uses, and growing tips
Healthcare Practitioners	Detailed herb profiles with scientific names, dosage info, and interactions
Educators / Researchers	Categorised plant data, filter by region/use, export or share profiles
Content Administrators	Add, edit, and manage herb entries, images, and categories
System Administrator	Manage users, roles, database backup, and system configuration

Non-Functional Requirements

Requirement	Specification
Page Load Time	< 2 seconds for all herb listing pages
Availability	99.5% uptime; hosted on cloud infrastructure
Data Security	JWT authentication + role-based access control
Scalability	Support 500+ herb entries and 200+ concurrent users

DAY 3 **Activity:** Objective Definition | **Deliverable:** Project Objectives | *Jun-03*

Project Objectives

Obj #	Objective Description
OBJ-01	Develop a comprehensive herbal plant database with scientific and common names
OBJ-02	Implement an interactive search and filter system by herb name, category, or region
OBJ-03	Display rich herb profiles with images, medicinal uses, and cultivation details
OBJ-04	Provide an admin panel for content management (CRUD operations on herb entries)
OBJ-05	Enable user registration, login, and bookmarking of favourite herbs
OBJ-06	Include a virtual garden feature allowing users to create personal herb collections
OBJ-07	Provide a quiz and learning module on herbal knowledge for educational purposes
OBJ-08	Ensure responsive design for desktop, tablet, and mobile access

SMART Goals

Criterion	Description
Specific	Build a herbal encyclopedia web app with search, profiles, and admin panel in 30 days
Measurable	All 8 objectives completed with 100+ herb entries and a working quiz module
Achievable	Technologies are well-documented; prior HTML/CSS/JS/MySQL skills available
Relevant	Directly addresses the gap in accessible, structured herbal knowledge platforms
Time-Bound	Complete by June 30, 2026 with full submission and live demo

DAY 4 **Activity:** User & Module Identification | **Deliverable:** Module List | *Jun-04*

User Roles

User Role	Description	Access Level
Guest / Visitor	Browse herbs, search, view profiles without login	Read-only
Registered User	Bookmark herbs, maintain personal garden, attempt quizzes	Basic
Content Manager	Add/edit/delete herb entries, images, and categories	Medium
System Administrator	Full system config, user management, database control	Full

Module List

Module #	Module Name	Description
M-01	User Management	Registration, Login, JWT auth, role-based access
M-02	Herb Database	Stores all herb entries with names, properties, images, and categories
M-03	Search & Filter Engine	Full-text search, category filter, region-based browsing
M-04	Herb Profile Viewer	Detailed page per herb: images, uses, dosage, growth tips
M-05	Virtual Garden	User's personal collection of bookmarked / grown herbs
M-06	Quiz & Learning Module	MCQ-based quizzes, score tracking, herbal knowledge tests
M-07	Admin / Content Panel	CRUD interface for herbs, categories, media, and user roles
M-08	Reporting & Analytics	Most-viewed herbs, quiz stats, user engagement reports

DAY 5 Activity: Use Case Diagram Preparation | Deliverable: CRUD APIs | Jun-05

Use Case Summary

Actor	Use Case
Guest	Browse herb catalogue / Search herbs / View herb profile
Registered User	Login / Bookmark herb / Add to virtual garden / Attempt quiz / View score
Content Manager	Login to Admin Panel / Add Herb / Edit Herb / Upload Image / Manage Categories
System Administrator	Manage Users and Roles / Configure System / View Reports / Backup DB
System (Search)	Auto-suggest herb names on keystroke / Apply filters dynamically
System (Quiz)	Auto-generate questions / Calculate score / Display leaderboard

CRUD API Matrix

Entity	Create	Read	Update	Delete
User	POST /api/users/register	GET /api/users/{id}	PUT /api/users/{id}	DELETE /api/users/{id}
Herb	POST /api/herbs	GET /api/herbs/{id}	PUT /api/herbs/{id}	DELETE /api/herbs/{id}
Category	POST /api/categories	GET /api/categories	PUT /api/categories/{id}	DELETE /api/categories/{id}
Bookmark	POST /api/bookmarks	GET /api/bookmarks/{uid}	—	DELETE /api/bookmarks/{id}
Quiz	POST /api/quiz	GET /api/quiz/{id}	PUT /api/quiz/{id}	—
Quiz Result	POST /api/results	GET /api/results/{uid}	—	—
Report	POST /api/reports	GET /api/reports/{id}	—	—

DAY 6

Activity: Database Requirement Analysis | Deliverable: Table List | Jun-06

Database Tables Identified

Table Name	Purpose	Key Columns
users	Registered user accounts and roles	user_id, name, email, role, password_hash
herbs	All herbal plant entries	herb_id, common_name, scientific_name, category_id, description, region
categories	Herb classification groups	category_id, name, description
herb_images	Images linked to herb entries	image_id, herb_id (FK), url, caption, uploaded_at
herb_properties	Medicinal properties per herb	property_id, herb_id (FK), use_type, description, dosage
bookmarks	User-saved favourite herbs	bookmark_id, user_id (FK), herb_id (FK), saved_at
quiz_questions	MCQ questions on herbs	question_id, herb_id (FK), question_text, options (JSON), correct_option
quiz_results	Quiz attempt scores per user	result_id, user_id (FK), score, total, attempted_at
system_logs	System activity and audit logs	log_id, action, user_id, timestamp, details

Database Design Decisions

- MySQL used for all structured relational data (users, herbs, bookmarks, quizzes)
- All tables follow 3NF (Third Normal Form) for minimal redundancy
- Auto-increment integers used as primary keys for performance
- Timestamps stored in UTC format for consistency across time zones
- Indexed columns: herb_id, category_id, user_id for fast query performance

DAY 7 **Activity:** ER Diagram Design | **Deliverable:** ER Diagram | Jun-07

Entity-Relationship Overview

The ER diagram captures all entities and their relationships. Core entities and attributes are listed below.

■ **HERBS**

Attribute	Type / Description
herb_id (PK)	INT, NOT NULL, AUTO_INCREMENT
common_name	VARCHAR(150), NOT NULL
scientific_name	VARCHAR(200), UNIQUE
category_id (FK)	INT → categories
region	VARCHAR(100)
description	TEXT
created_at	TIMESTAMP DEFAULT NOW()

Relationship: One CATEGORY has many HERBS (1:N) | One HERB has many IMAGES and PROPERTIES (1:N)

■ **HERB_PROPERTIES**

Attribute	Type / Description
property_id (PK)	INT, NOT NULL, AUTO_INCREMENT
herb_id (FK)	INT → herbs
use_type	ENUM (medicinal, culinary, aromatic, cosmetic)
description	TEXT, NOT NULL
dosage	VARCHAR(200)

Relationship: One HERB has many PROPERTIES (1:N)

■ **BOOKMARKS**

Attribute	Type / Description
bookmark_id (PK)	INT, NOT NULL, AUTO_INCREMENT
user_id (FK)	INT → users
herb_id (FK)	INT → herbs
saved_at	TIMESTAMP, NOT NULL

Relationship: One USER has many BOOKMARKS (1:N) | Many-to-many USERS ↔ HERBS via bookmarks

■ **QUIZ_QUESTIONS**

Attribute	Type / Description
question_id (PK)	INT, NOT NULL, AUTO_INCREMENT
herb_id (FK)	INT → herbs
question_text	TEXT, NOT NULL
options	JSON — array of 4 choices

Attribute	Type / Description
correct_option	TINYINT — index 0–3

Relationship: One HERB can have many QUIZ_QUESTIONS (1:N)

DAY 8 **Activity:** Database Schema Creation | **Deliverable:** SQL Schema | *Jun-08*

SQL Schema Overview

The MySQL schema was created with all tables, constraints, indexes, and foreign key relationships as defined below.

Schema: Relationships Map

From Table	To Table	Relationship	Key
categories	herbs	1:N — category classifies herbs	category_id → category_id
herbs	herb_images	1:N — herb has images	herb_id → herb_id
herbs	herb_properties	1:N — herb has properties	herb_id → herb_id
herbs	quiz_questions	1:N — herb has questions	herb_id → herb_id
users	bookmarks	1:N — user saves bookmarks	user_id → user_id
users	quiz_results	1:N — user has quiz results	user_id → user_id
quiz_questions	quiz_results	1:N — questions generate results	question_id → question_id

Index Strategy

Table	Index Column(s)	Reason
users	email	Fast login and duplicate-check lookup
herbs	category_id, common_name	Category filtering and name search
herb_properties	herb_id, use_type	Medicinal-use filtering per herb
bookmarks	user_id, herb_id	Quick retrieval of user's saved herbs
quiz_results	user_id, attempted_at	Leaderboard and history queries

DAY 9 **Activity:** UI Wireframe Design | **Deliverable:** Page Layouts | *Jun-09***Wireframe Page List**

Page #	Page Name	User Role	Key Elements
P-01	Home / Landing Page	All	Hero banner, search bar, featured herbs, category grid
P-02	Herb Catalogue	All	Herb cards grid, filter sidebar, pagination, sort options
P-03	Herb Profile Page	All	Image gallery, properties table, uses, dosage, bookmark button
P-04	Virtual Garden	Registered User	User's saved herbs, garden view toggle, notes field
P-05	Quiz Module	Registered User	MCQ questions, timer, progress bar, score display
P-06	Leaderboard	All	Top quiz scores, rank, username, date
P-07	Admin Dashboard	Admin/Manager	Stats cards, recent activity, quick-add button
P-08	Herb Management	Admin/Manager	Herb table, add/edit/delete, image upload, category assign
P-09	User Management	Admin	User list, role editor, activate/deactivate, search/filter

UI Design Principles

- Responsive design for desktop, tablet, and mobile
- Natural green and cream color palette matching the herbal/botanical theme
- Card-based herb browsing with high-quality images as the primary UI element
- Minimum 14px body text size with high contrast for readability
- React.js with Tailwind CSS for rapid and consistent UI development

DAY 10

Activity: Login & Dashboard UI Design | Deliverable: UI Screens | Jun-10

Login Screen Design

Component	Description
Logo & Title	Leaf logo with 'Virtual Herbal Garden' heading in dark green
Email Input	Placeholder: 'Enter your email', with format validation
Password Input	Masked input field with show/hide toggle icon
Role Selector	Dropdown: Guest / Registered User / Content Manager / Admin
Login Button	Primary green button, disabled until form is valid
Register Link	'New here? Create an account' — link below the login button
Error Messages	Red inline validation messages for invalid or missing inputs

Main Dashboard / Home UI Components

Component	Description
Navbar	Logo, Search bar, Category dropdown, Login/Profile button
Hero Banner	Full-width botanical image with 'Explore Herbs' call-to-action button
Featured Herbs	Horizontal scroll row showing 6 featured herb cards with images
Category Grid	Icon tiles: Medicinal / Culinary / Aromatic / Cosmetic / Regional
Stats Bar	Total herbs, total categories, registered users, quizzes taken
Footer	About, Contact, Social links, Newsletter subscription field

Herb Profile Page — Key Components

- Large image gallery with a thumbnail strip below
- Common name (h1 heading), Scientific name (italic subtitle), Category badge
- Tabbed sections: Overview | Medicinal Uses | Cultivation | Recipes
- Properties table: use type, dosage, and preparation method
- Bookmark / Add to Garden buttons (visible only to registered users)
- Related herbs carousel shown at the bottom of the page

DAY 11 **Activity:** Navigation & Form Design | **Deliverable:** UI Prototype | *Jun-11***Navigation Structure**

Navigation Item	Guest	Registered User	Admin / Manager
Home	Yes	Yes	Yes
Herb Catalogue	Yes	Yes	Yes
Herb Profile	View Only	Full Access	Full Access
Virtual Garden	No	Yes	Yes
Quiz Module	No	Yes	Yes
Admin Dashboard	No	No	Full CRUD
User Management	No	No	Yes
Profile / Settings	No	Yes	Yes

Key Forms Designed

Form Name	Fields	Validation
User Registration	Name, Email, Phone, Password, Confirm Password	Email format, phone 10 digits, password strength
Add Herb Entry	Common Name, Scientific Name, Category, Region, Description, Image Upload	Required fields, valid category, image < 2MB
Herb Property Form	Use Type, Description, Dosage, Preparation Method	Use type required; dosage must be numeric if entered
Quiz Question Form	Question Text, Option A/B/C/D, Correct Option, Herb Link	All 4 options required; correct option selection mandatory

Prototype Flow

- Figma prototype: All 9 screens connected with click-through interactions
- Guest flow: Home → Browse Catalogue → View Herb Profile → Register
- User flow: Login → Virtual Garden → Add Herb → Take Quiz → View Score
- Admin flow: Login → Dashboard → Manage Herbs → Manage Users → View Reports

DAY 12 **Activity:** Design Review | **Deliverable:** Design Approval | *Jun-12*

Design Review Checklist

Review Area	Status	Comments
Project Title	Approved	Title is clear, specific, and educationally relevant
Problem Statement	Approved	Well-defined scope with real-world significance
Objectives (8)	Approved	All SMART objectives met and clearly defined
Module List (8)	Approved	All modules accounted for and described
Use Cases	Approved	Covers all actors and interactions comprehensively
CRUD APIs	Approved	REST conventions followed correctly
Database Tables (9)	Approved	3NF normalised, indexes defined
ER Diagram	Approved	All entity relationships correctly mapped
SQL Schema	Approved	Constraints and foreign keys properly defined
UI Wireframes	Approved	9 screens fully designed for all user roles
UI Screens	Approved	Login and Dashboard components complete
UI Prototype	Minor Revision	Mobile card layout needs padding fix on small screens
Overall Design	Approved	Ready to proceed to development phase

Faculty Review Feedback

1. Strong project with clear educational and public-health applicability. Herb database design is thorough.
2. Ensure the search engine includes synonym support (e.g., 'Tulsi' and 'Holy Basil' returning the same result).
3. Quiz module should include difficulty levels — Easy, Medium, and Hard — for progressive learning.
4. Overall: Design phase is well-executed and technically sound. Approved for development phase.

DAY 13 **Activity:** Frontend Environment Setup | **Deliverable:** React Project Setup | *Jun-13***Frontend Tools & Packages**

Tool / Package	Version	Purpose
Node.js	v20.x LTS	JavaScript runtime environment
npm	v10.x	Package manager
React.js	v18.2	Frontend framework
Vite	v5.x	Build tool (faster than Create React App)
Tailwind CSS	v3.4	Utility-first CSS framework
React Router DOM	v6.x	Client-side routing and navigation
Axios	v1.6	HTTP client for REST API calls
React Query	v5.x	Server state management and caching
Swiper.js	v11.x	Image gallery and herb carousel slider
React Toastify	v10.x	Notification toasts for user actions
ESLint + Prettier	Latest	Code quality and auto-formatting

Project Folder Structure

```
herbal-garden-frontend/  
  ■■■■ src/  
    ■ ■■■■ components/ # HerbCard, SearchBar, QuizCard, Navbar  
    ■ ■■■■ pages/ # Home, Catalogue, HerbProfile, Garden, Quiz  
    ■ ■■■■ services/ # herbService.js, quizService.js, authService.js  
    ■ ■■■■ context/ # AuthContext.jsx, GardenContext.jsx  
    ■ ■■■■ hooks/ # useHerbSearch.js, useQuizTimer.js  
    ■ ■■■■ utils/ # formatters.js, validators.js  
    ■ ■■■■ App.jsx # Root component with routes  
    ■■■■ public/ # favicon, hero images, icons  
    ■■■■ package.json # Dependencies manifest
```

Environment Variables (.env)

- VITE_API_BASE_URL=http://localhost:5000/api
- VITE_MAPS_API_KEY=[Google Maps API Key]
- VITE_QUIZ_TIMER_SECONDS=30

Days 1 – 13 | Progress Summary

Day	Date	Planned Activity	Expected Deliverable
01	Jun-01	Project Title Finalization	Approved Project Title
02	Jun-02	Requirement Gathering	Problem Statement
03	Jun-03	Objective Definition	Project Objectives
04	Jun-04	User & Module Identification	Module List
05	Jun-05	Use Case Diagram Preparation	CRUD APIs
06	Jun-06	Database Requirement Analysis	Table List
07	Jun-07	ER Diagram Design	ER Diagram
08	Jun-08	Database Schema Creation	SQL Schema
09	Jun-09	UI Wireframe Design	Page Layouts
10	Jun-10	Login & Dashboard UI Design	UI Screens
11	Jun-11	Navigation & Form Design	UI Prototype
12	Jun-12	Design Review	Design Approval
13	Jun-13	Frontend Environment Setup	React Project Setup